

Challenges in Pulsed-Field Gradient Nuclear Magnetic Resonance on Magnetically Heterogeneous Interfaces: Sequence and Field-Dependent Apparent Diffusion Coefficients

Kee Sung Han* and Pasquale F. Pulvio

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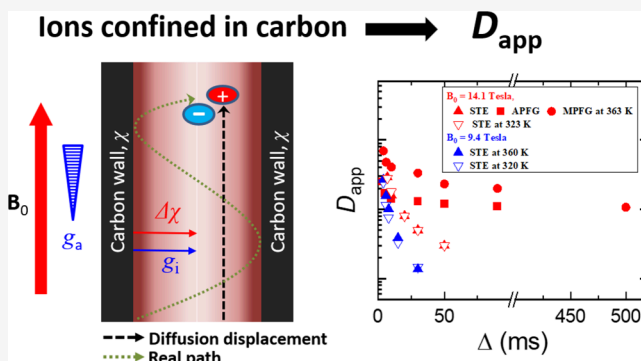
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ABSTRACT: It is well known that the internal gradient (g_i) that exists within pores haunts the diffusion coefficient (D) as measured by the pulsed-field gradient (PFG) nuclear magnetic resonance (NMR). Several PFG-NMR methods developed to determine accurate D were not successful. Then, the steady-state diffusion coefficient ($D_{app,\infty}$) for the cation $[C_4mim]^+$ of $[C_4mim][Tf_2N]$ [1-butyl-3-methylimidazolium][bis(trifluoromethylsulfonyl)imide] ionic liquid confined in ordered mesoporous carbon (OMC) was determined by comparing $D_{app,\infty}$ obtained from 1H PFG-NMR performed with three different stimulated echo sequences: STE, alternating pulsed-field gradient (APFG), and magic pulsed-field gradient (MPFG) under the two external magnetic field strengths, $B_0 = 9.4$ and 14.1 T. The measured $D_{app,\infty}$, which is an order of magnitude smaller than D of bulk $[C_4mim][Tf_2N]$, is in good agreement between APFG and MPFG in both $B_0 = 9.4$ and 14.1 T. However, the strong g_i artifact, which caused apparent diffusion coefficient (D_{app}), depending strongly and weakly on B_0 and temperature, respectively, in diffusion time-dependent D_{app} , $D_{app}(\Delta)$, obtained from a sequence with monopolar gradients (STE), was suppressed by using sequences employing bipolar gradients (APFG and MPFG) in the region of steady-state diffusion. However, the incompletely suppressed g_i artifact resulting in the different behaviors of the early part of $D_{app}(\Delta)$ between the sequences leads to $\langle z(t)^2 \rangle = 2Dt^\alpha$, where $\alpha = 0.5$ and 1 for one-dimensional single-file diffusion and 3D bulk diffusion, respectively. The above observations clearly show that the diffusion behavior of ions/molecules within the pores and pore structure, such as the surface-to-volume ratio $\left(D_{app}(\Delta) = D_0 \left[1 - \frac{4}{9\sqrt{\pi}} \frac{S}{V} \sqrt{D_0 \Delta} \right] \right)$ and tortuosity ($T \equiv D_0/D_{app,\infty}$), is possible to be misunderstood, especially in systems with a non-negligible g_i . This work demonstrates that it may be necessary to test several PFG sequences under multiple external magnetic fields for the correct determination of the diffusion behavior of ions or molecules in the pores with a larger internal gradient, g_i .



INTRODUCTION

Pulsed-field gradient nuclear magnetic resonance (PFG-NMR) determines the diffusion coefficient (D) of molecules or ions using an echo signal attenuation due to applying a pulsed gradient. It is challenging to apply PFG-NMR to (magnetically) heterogeneous samples such as viscous liquids, fluids at fluid/solid interfaces, and ions in solid-state materials such as solid-state lithium-ion conductors due to additional attenuation of echo signals induced by the internal gradient (g_i) and effective gradient (i.e., cross term) arising from the interference between internal (g_i) and applying (g_a) gradients.^{1–6} In addition, the nuclear relaxation time (T_1 and/or T_2) in these materials is quite fast relative to the diffusion time, Δ , and therefore precludes detecting echo signals after applying Δ in PFG-NMR (Scheme 1). For samples with short relaxation times, it may be possible to use a simple spin-echo (SE) PFG sequence, which is composed

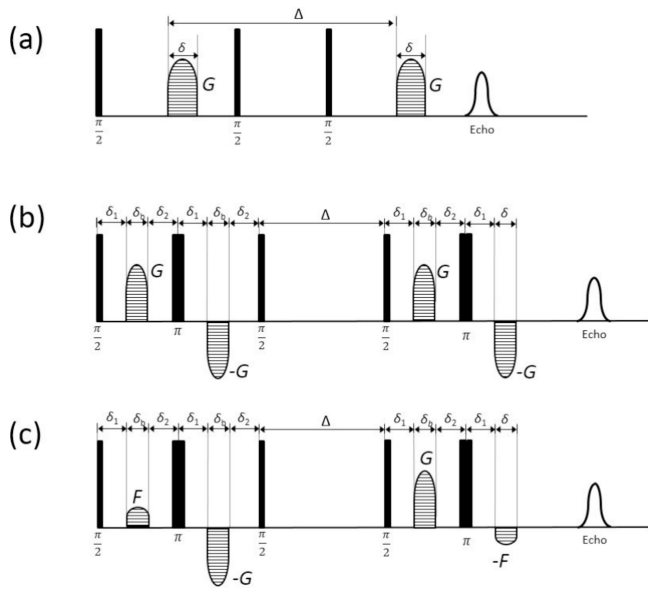
of two radio frequency (rf) and two gradient pulses. However, this simple PFG sequence may not suppress the artifact arising from the internal gradient (g_i).⁷ For the suppression of this artifact, this simple SE sequence was modified to a stimulated echo sequence composed of mono or bipolar gradients, as shown in Scheme 1.⁸ Data obtained utilizing the stimulated echo (STE) with monopolar gradients (Scheme 1a) revealed that this pulse sequence can suppress the internal gradient artifacts of

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Scheme 1. Stimulated Echo Pulsed-Field Gradient NMR Sequences for Diffusion Measurements with (a) Monopolar Gradients (STE) and (b) Symmetric and (c) Asymmetric (G/F Ratio ≈ -6.7) Bipolar Gradients, the So-Called Alternating Pulsed-Field Gradient (APFG) and Magic Pulsed-Field Gradient (MPFG), Respectively



viscous liquids⁷ and liquid/solid interfaces containing a solid phase with weak magnetic susceptibility (χ).⁹ However, STE with a monopolar gradient may not be perfect for the spatially nonconstant background gradient. Then, the stimulated echo sequences employing bipolar gradients (Scheme 1b,c) were originally developed for the suppression of the internal gradient artifact due to the nonconstant internal (background) gradient.^{1,3} These two sequences use a 13-interval stimulated echo with symmetric and asymmetric bipolar pulsed gradient ratios, alternating pulsed-field gradient (APFG) and magic pulsed-field gradient (MPFG) ratios,³ respectively. These modified stimulated echo sequences restrict the minimum Δ that can be used during the experiment due to a greater number of rf and gradient pulses between the encoding and decoding gradient pulses. Therefore, it is usually not feasible to use modified STE sequences for samples with short relaxation times ($\leq 4-5$ ms). Then, the STE sequence with a monopolar gradient (Scheme 1a) was preferable for samples with short relaxation times.^{10,11}

In this study, room temperature ionic liquid (RTIL), $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ confined in ordered mesoporous carbon (OMC), was selected as a model system because RTIL has received significant attention as potential candidates for electrolyte materials in electrochemical devices such as batteries, supercapacitors, and fuel cells,¹²⁻¹⁴ and carbon is widely used as electrode material in most electrochemical devices. The structure and dynamics of the IL at the interface influence the chemical reactions that occur in these devices.^{13,15} For these applications, it requires understanding the transport properties of ILs and their variation at liquid/solid interfaces due to the interaction between the ILs and solid surfaces. RTILs are well known for their thermal and chemical stability, reasonable ionic conductivity, and wide electrochemical window. However, these properties are modified by the host materials due to the IL-surface interactions. Many studies have been performed on the

properties of IL at interfaces, such as the conformation and dynamics of ILs and on the device performances.^{13,16-22} The results showed that IL displayed a dynamic distribution in the pore resulting from decreasing both translational and rotational motion of ILs on the pore walls.^{9,15} In addition, device performance is closely related to the structure and size of the pores and the chemical and thermal stability and (translational) mobility of ILs.^{21,23-25} However, it is not easy to use PFG-NMR for the determination of D of ILs confined in carbon pores due to the internal gradient artifact induced by inhomogeneous magnetic susceptibility along the pore wall to those portions of the pore that are not affected by changing magnetic susceptibility. The methods that successfully measured the diffusion coefficient for systems with Fe_2O_3 -doped MgO paste⁴ and monosized glass beads immersed in water failed to find a reasonable diffusion coefficient for the RTIL-OMC system tested in this study.⁵ In contrast, accurate diffusion coefficients for the $[\text{C}_4\text{mim}]^+$ cation of $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ confined in an OMC were determined by comparing the diffusion coefficients at the steady state from three different stimulated echo sequences: STE, APFG, and MPFG (Scheme 1), under magnetic field strengths of 9.4 and 14.1 T by using ^1H PFG-NMR. The results were compared to the D values calculated from molecular dynamics (MD) simulations.¹⁵ The internal gradient artifact, which was observed clearly in STE, was suppressed by APFG and MPFG in steady-state diffusion. Also, the behaviors of apparent diffusion coefficients $D_{\text{app}}(\Delta)$ obtained as a function of Δ varied with applying PFG-NMR sequences, probably due to the different efficiencies for the suppression of the internal gradient artifact.

EXPERIMENTAL SECTION

Sample Preparations

Ordered mesoporous carbon, OMC-80/850, was prepared by the self-assembly of resorcinol and formaldehyde in the presence of triblock copolymer F127. The gel obtained from this mixture was cured overnight at 80 °C for 24 h. The sample was carbonized by heating in flowing nitrogen at 2 °C/min to 850 °C and held at this temperature for 2 h. The OMC has a single-point pore volume of 0.59 cm^3/g , a micropore volume of 0.18 cm^3/g , pore width of 7.3 nm, and a specific surface area of 673 m^2/g . The details of the measurement techniques for the characterization are in ref 26. The IL $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ was synthesized from methylimidazole in two steps, according to literature procedures.²⁷ The amount of $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ required to fill the pore volume was calculated using the carbon pore volume and $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ density of 1.43 $\text{g}\cdot\text{cm}^{-3}$.²² A $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ /ethanol solution was added to the carbon and stirred for 1 h at room temperature. The ethanol was removed in a weak vacuum (~ 20 mmHg) at 80 °C for 24 h.

NMR and PFG-NMR Experiments

The D for the $[\text{C}_4\text{mim}]^+$ cation of the $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ ionic liquid confined in ordered mesoporous carbon (OMC) was measured by ^1H PFG-NMR with three different pulse sequences represented in Scheme 1. The ^1H PFG-echo profile and echo intensity as a function of gradient strength, measured at 303 K with $\Delta = 90$ ms using APFG, are presented in Figure S1, indicating behavior consistent with normal diffusion measurements. Previous reports showed that MPFG can suppress the artifact due to internal gradient (g_i) and the cross term between internal and external gradients, which cannot be perfectly removed by APFG.³ For the MPFG experiment, the gradient ratio ($\eta = G/F$) of -6.7114 was used and the gradient pulses (δ_b) were applied at the center of the interval ($\delta_1 = \delta_2$) between the two rf pulses, τ , as shown in Scheme 1c. A 600 MHz NMR spectrometer (Agilent, USA) equipped with a 5 mm z-gradient probe (Doty Scientific, USA) with a maximum gradient strength of ~ 31 T/m was used. The echo height, $S(g)$, recorded as a function of gradient strength, g , was fitted with the Stejskal-Tanner

equation, $S(g) = S(0)\exp[-D(\gamma g \Delta)^2(\Delta - \delta/3)]$ for STE and APFG,⁸ where $S(g)$ and $S(0)$ are the echo heights at the gradient strengths of g and 0, respectively, D is the diffusion coefficient, γ is the ^1H gyromagnetic ratio, Δ is the time interval between the two pulse gradients, i.e., diffusion delay, and δ is the gradient length. For the MPFG measurement, the diffusion coefficient was calculated with $S(g) = S(0)\exp(-D\gamma^2 A_p)$, where $A_p = (\delta_b F)^2[(\Delta + \tau)(1 + \eta)^2 + 2\tau - 1/3\delta_b(1 + \eta^2)]$.³ Here, δ_b corresponds to $\delta/2$ in STE sequences because the MPFG uses a pair of bipolar gradients instead of a monopolar gradient. The gradient strength was varied with 15 equal steps, and the maximum gradient strength was chosen according to the echo height at the maximum gradient between 10 and 30 T/m. The 90° pulse length and gradient duration δ were 8 μs and 1.5 ms, respectively. Apparent diffusion coefficients, $D_{\text{app}}(\Delta)$, as a function of Δ , were obtained from $\Delta = 3$ to 500 ms. The steady-state diffusion coefficients, $D_{\text{app},\infty}$, were set equal to the D_{app} obtained at the longest Δ , and $D_{\text{app},0}$ was extrapolated to $\Delta = 0$. To monitor the external field effect on the internal gradient, the measurements with an STE sequence were also performed on a 400 MHz Bruker Avance NMR spectrometer equipped with a 5 mm z-gradient probe, which has a maximum gradient strength of ~ 29 T/m (Diff60, Bruker, Germany) at a ^1H Larmor frequency of 400.16 MHz in the temperature range from 300 to 360 K. The gradient strength was varied with 16 equal steps, and the 90° pulse length and δ were 7 μs and 3 ms, respectively. Δ was varied from 4 to 30 ms. The bulk diffusion coefficients of $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ were measured with a 5 mm z-gradient probe, which has a maximum gradient of 54.4 G/cm (0.0544 T/m). The details of PFG-NMR experiments performed on a 400 MHz spectrometer can be found in ref 9.

RESULTS AND DISCUSSION

The ^1H NMR spectra of the bulk and OMC-confined $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ are plotted in Figure 1. The peaks of the ^1H

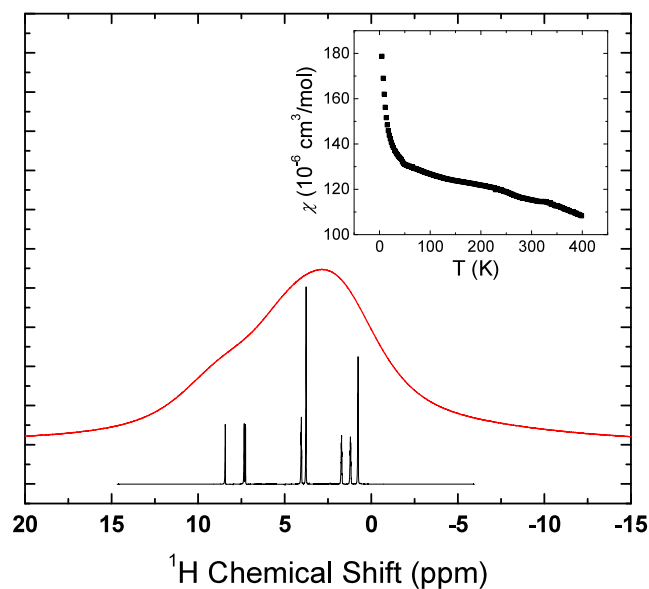
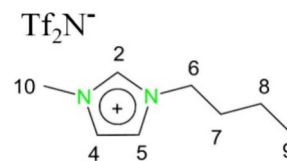


Figure 1. ^1H NMR spectra for the $[\text{C}_4\text{mim}]^+$ cation of $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ confined in OMC (top) and as a neat liquid (bottom) at 9.4 T, measured at 300 K. The inset shows the magnetic susceptibility of OMC used for the confinement of $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$.

NMR spectrum obtained from the bulk $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ can be assigned to H2, H5, H4, H6, H10, H7, H8, and H9 from left to right (Scheme 2). The H5 and H4 are too close to be resolved at this x-axis scale. The peaks on the OMC-confined $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ are much broader than on the bulk $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ due to the magnetic field inhomogeneity (ΔB) in the carbon pore, which is roughly estimated to be 0.9 gauss from $T_2 \approx 5$ ms and

Scheme 2. Chemical Structure of the $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ Ionic Liquid^a



^aThe numerals label the proton resonances.

line width ≈ 4 kHz (~ 10 ppm at 9.4 T) at 300 K. This is 3 times larger than the ΔB in the silica mesopore.⁹ This larger ΔB , which arises due to the magnetic susceptibility differences between the wall of host materials and the volume spaces of the pore,²⁸ might be the source of an internal gradient (g_i) that affects the diffusion measurements. The magnetic susceptibility, χ (defined below), of the OMC (inset, Figure 1) exhibited typical paramagnetic behavior with a strong positive susceptibility decreasing with temperature. Though the temperature dependence changes abruptly at approximately 50 K, χ remained large (1.16×10^{-4} cm³/mol) and the slope (8×10^{-8} cm³/mol·K) over the temperature range of this study was steep relative to that of mesoporous silica, KIT-6 (Figure S2). The temperature dependency of the carbon χ may result in a variable internal gradient, which will manifest itself in the cross-term contribution to the diffusion coefficient¹ and result in temperature-independent and external magnetic field-dependent $D_{\text{app}}(\Delta)$, as shown in Figure 2. The magnetic susceptibility is defined as χ

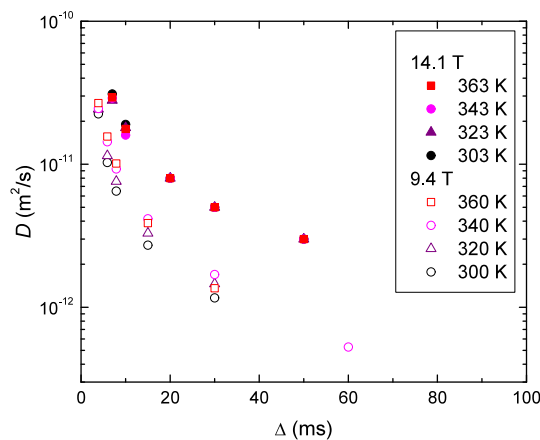


Figure 2. $D_{\text{app}}(\Delta)$ of the $[\text{C}_4\text{mim}]^+$ cation of $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ confined in the OMC obtained by using STE (monopolar gradient stimulated echo, Scheme 1a) at various temperatures under external magnetic field strengths of 9.4 and 14.1 T. While the decrease in $D_{\text{app}}(\Delta)$ with the increase in Δ is typical of diffusion under confinement, the temperature-independent and external magnetic field-dependent behavior of $D_{\text{app}}(\Delta)$ strongly suggest that internal gradient artifacts influence $D_{\text{app}}(\Delta)$.

$= \Delta \mathbf{M} / \Delta \mathbf{H}$, where $\mathbf{M}$ is the magnetization and $\mathbf{H}$ is the applied magnetic field intensity that generates the magnetic flux density, $\mathbf{B}$, i.e., magnetic field, in the specimen according to the equation $\mathbf{B} = \mathbf{H} + 4\pi \mathbf{M} = (1 + 4\pi \chi) \mathbf{H}$. A magnetic flux density inhomogeneity ($\Delta \mathbf{B}$) arises on the inside of the pore along the pore wall moving toward the pore volume, resulting in a magnetic flux density gradient, referred to as the internal (magnetic) field gradients, g_i . To obtain differential g_i by variation of $\mathbf{B}$: $\Delta \mathbf{B} = (1 + 4\pi \chi) \Delta \mathbf{H}$ of the OMC, $\mathbf{H}$ (9.4 or 14, 1 T in this study) was varied, resulting in the external magnetic field-

dependent $D_{\text{app}}(\Delta)$, as shown in Figure 2 for the $D_{\text{app}}(\Delta)$ obtained using STE (Scheme 1a). It shows that $D_{\text{app}}(\Delta)$ decreased by 2 and 1 orders of magnitude over $\Delta = 60$ ms in 9.4 and 11.4 T, respectively. This suggests that the negligible effect of temperature on $D_{\text{app}}(\Delta)$ may be related to temperature varying χ , which can modify $B:\Delta B = (1 + 4\pi\Delta\chi)H$ of the OMC with temperature.

Based on these observations, the $D_{\text{app}}(\Delta)$ values spanning 2 orders of magnitude within a short Δ range are thought to be the result of the internal gradient artifacts.^{4–6} In contrast to OMC, mesoporous silica (KIT-6) exhibited weak diamagnetic susceptibility ($-1.8 \times 10^{-5} \text{ cm}^3/\text{mol}\cdot\text{K}$) with a negligible temperature dependence ($-1 \times 10^{-8} \text{ cm}^3/\text{mol}\cdot\text{K}$) (Figure S2), which did not affect the diffusion measurements of its guest ionic liquid $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$.⁹ A comparison of D of ILs confined in OMC and KIT-6 indicates that STE sequences may not be able to suppress the internal gradient artifacts arising from solid/liquid interfaces with a large magnetic field inhomogeneity, even though it works well for viscous liquids⁷ and liquids confined in pores with weak magnetic susceptibilities such as mesoporous silica,⁹ which resulted in the same $D_{\text{app}}(\Delta)$ of the $[\text{C}_4\text{mim}]^+$ cation of $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ confined in mesoporous silica from STE and APFG sequences (Figure S3a). In contrast to STE, APFG and MPFG showed that $D_{\text{app}}(\Delta)$ varied as a function of temperature, as shown in Figure 3. In addition, both APFG and MPFG yielded the same $D_{\text{app},\infty}$ while the $D_{\text{app}}(\Delta)$ values at the early range of Δ were not the same. D_{app} depends less on the temperature in the early range of Δ (Figure 3), while all $D_{\text{app},0}$

values merged to similar values at all measured temperatures (Figure S4). These findings strongly suggest that the internal gradient artifact is evident in the early stage of Δ and suppressed when utilizing a sufficiently long Δ for both APFG and MPFG. Based on the different behaviors of $D_{\text{app}}(\Delta)$ from APFG and MPFG, it was determined that the efficiency of g_i artifact suppression is different between these two sequences. $D_{\text{app}}(\Delta)$ obtained by MPFG is larger than that obtained by APFG at the early range of Δ (10–40 ms) and is pronounced at lower temperatures. Also, it requires longer Δ to reach steady-state diffusion ($D_{\text{app},\infty}$) in MPFG. At 260 K and $\Delta = 40$ ms, $D_{\text{app}} \approx 1.5 \times 10^{-12}$ and $5.2 \times 10^{-12} \text{ m}^2/\text{s}$ in APFG and MPFG, respectively. These values are larger or similar to the bulk diffusion coefficient $D_0 \approx 2.4 \times 10^{-12} \text{ m}^2/\text{s}$ for the $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ IL. Also, lower temperature $D_{\text{app},0}$ values are larger than the diffusion coefficient (D_0) of bulk IL (Figure S4).

To determine whether this larger $D_{\text{app},0}$ and rapid decrement of $D_{\text{app}}(\Delta)$ in a short-range of Δ are reasonable in this system, the surface area-to-pore volume (S/V_p) ratio was calculated from $D_{\text{app}}(\Delta)$ and compared to that calculated from the measured volume ($0.59 \text{ cm}^3/\text{g}$) and surface area ($673 \text{ m}^2/\text{g}$). These values were compared to the corresponding values obtained from silica (KIT-6).⁹ Usually, $D_{\text{app}}(\Delta)$ is closely related to the geometry of the pore.^{29,30} The slope of the initial region of $D_{\text{app}}(\Delta)$ is related to the S/V_p ratio, which can be calculated from the time-dependent diffusion coefficient, $D_{\text{app}}(\Delta) = D_0 \left[1 - \frac{4}{9\sqrt{\pi}} \frac{S}{V} \sqrt{D_0 \Delta} \right]$. The S/V_p of OMC calculated from the measured pore volume ($0.59 \text{ cm}^3/\text{g}$) and surface area ($673 \text{ m}^2/\text{g}$) is ~ 1.6 times larger than that of mesoporous silica with $V_p = 1.11 \text{ cm}^3/\text{g}$ and $S = 783 \text{ m}^2/\text{g}$.⁹ However, the S/V_p estimated from $D_{\text{app}}(\Delta)$ of APFG at 360 K for OMC and silica showed that it is ~ 13 times larger in OMC (Table S1). These findings imply that the larger decrement of $D_{\text{app}}(\Delta)$ in the OMC is due to a g_i artifact. The tortuosity $T = D_{\text{app},0}/D_{\text{app},\infty}$ estimated from $D_{\text{app}}(\Delta)$ obtained by MPFG is 2–7 times larger than that from APFG. The S/V_p appears to be larger in MPFG than in APFG based on the decrement of $D_{\text{app}}(\Delta)$ in an early stage of Δ . This implies that the internal gradient artifact is suppressed less efficiently by MPFG during the early stages of Δ while suppressing the g_i artifact completely at long Δ times, even though Δ is ~ 5 times longer than that of APFG.

The $D_{\text{app}}(\Delta)$ is the same between APFG and MPFG when $\Delta = 90$ and 500 ms, respectively. It is well known that the D_{app} under the internal gradient is influenced by the distribution of internal gradient across the sample, applied gradient strength (g_a), and diffusion length.²⁹ In this case, the internal gradient effect was suppressed when the diffusion length, $\sqrt{D\Delta}$, is larger than $\sim 1.5(\pm 0.5) \times 10^{-6} \text{ m}$ at $\Delta \approx 500$ ms in the MPFG and $\sim 6.5(\pm 2.6) \times 10^{-7} \text{ m}$ at $\Delta \approx 90$ ms in the APFG, as shown in Table S2. The $\sqrt{D\Delta}$ at $D_{\text{app},\infty}$ is the length scale required to overcome the internal gradient artifact, and it is twice as large in MPFG as in APFG. These behaviors may be due to the weaker applied gradient strength of MPFG. The bipolar gradient pairs in MPFG have the ratio $|G|/|F| \sim 6.7$, which corresponds to 57% of the strength ($g_a = |G| + |F|$) of a symmetric bipolar gradient ($|G|/|F| = 1$) when the strength G is the same in both APFG and MPFG. It is well known that the internal gradient and the cross term of internal and applied gradients interfere with the echo height according to the strength of applied gradients.¹ The weaker g_a may result in larger g_i interference due to the contribution of the cross term, $g_i \cdot g_a$.¹ The maximum gradients

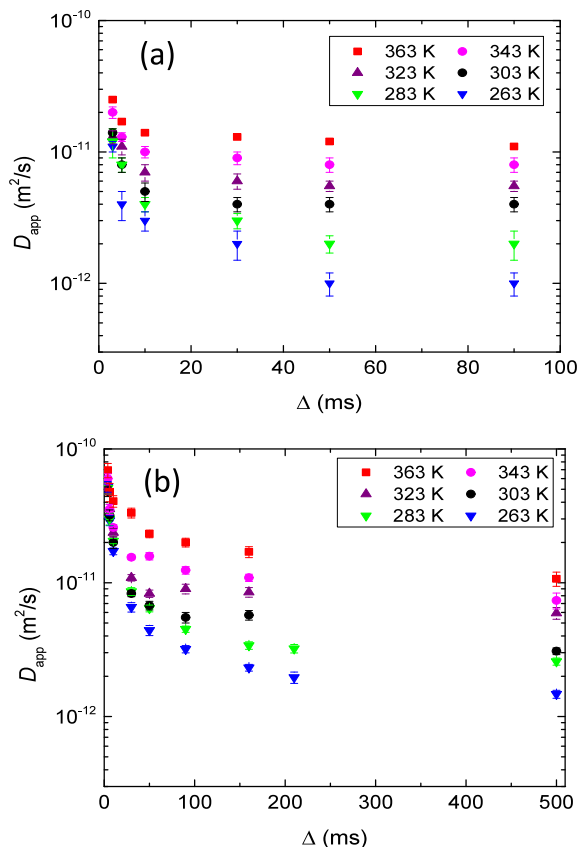


Figure 3. Apparent diffusion coefficients, $D_{\text{app}}(\Delta)$ of $[\text{C}_4\text{mim}]^+[\text{Tf}_2\text{N}]^-$ confined in the OMC as a function of diffusion time, Δ , obtained by ^1H PFG-NMR using (a) APFG and (b) MPFG at 14.1 T.

used in APFG at $\Delta = 90$ ms and MPFG at $\Delta = 500$ ms at 363 K were 1400 and 400 G/cm, respectively. This shows that it is preferable to use APFG with a reasonably higher gradient strength for suppressing internal gradient artifacts, and MPFG could be a second choice if the instrument could not generate a sufficiently high external gradient g_a . Meanwhile, there is still a possibility of fast diffusion of $[C_4mim]^+$ of $[C_4mim][Tf_2N]$ as reported by quasi-elastic neutron scattering (QENS) measurement.³¹ The fast diffusion of ions/molecules in the pores relative to those in the bulk liquid state, as we observed at lower temperatures within the pore of OMC, is possible under the condition of single-file diffusion.^{32,33} This possibility can be tested with $D_{app}(\Delta)$ using the relationship between the mean-squared displacement ($\langle z(t)^2 \rangle$) and diffusion time (t): $\langle z(t)^2 \rangle = 2Dt^\alpha$, where $\alpha = 0.5$ and 1 in a 1D single-file diffusion and 3D bulk diffusion, respectively.^{32,33} It showed that the MSD calculated using $D_{app}(\Delta)$ was proportional to diffusion time with orders of 0.63 and 0.85 from MPFG and APFG at 363 K, respectively (Figure 4), whereas these values would be identical

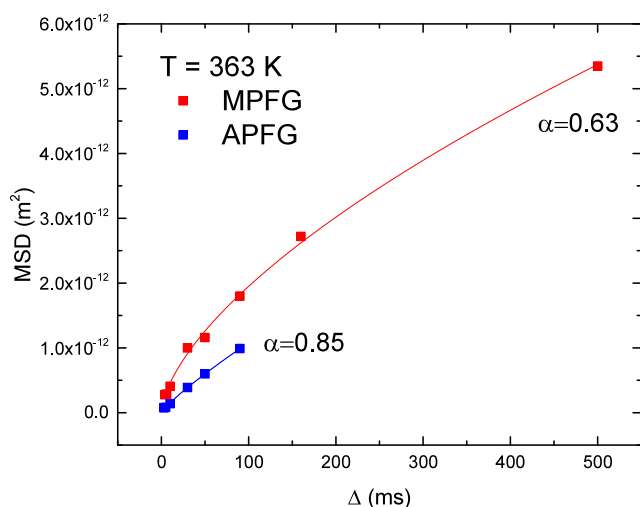


Figure 4. Mean-squared displacement (MSD) of $[C_4mim]^+$ of $[C_4mim][Tf_2N]$ confined in mesoporous carbon as a function of Δ calculated from $D_{app}(\Delta)$ at 363 K and 14.1 T.

in the absence of g_i -related artifacts in $D_{app}(\Delta)$. The origin of the smaller α value from MPFG is a consequence of both a larger decrement of $D_{app}(\Delta)$ at the early stage of Δ and longer Δ to reach steady-state diffusion due to the larger effective g_i artifact resulting from the weaker g_a . In the same manner, the α value being smaller than unity in APFG is also attributed to the g_i artifact at the early stage of Δ . If the estimation is performed without data from the early part of Δ , then the α value is close to unity (Figure S5). Therefore, it is concluded that the diffusion of the $[C_4mim]^+$ cation of $[C_4mim][Tf_2N]$ IL confined in OMC is consistent with 3D bulk diffusion as similar to those observed within silica pores, which showed $\alpha = 1$ (Figure S5). The changes to the effective g_i artifact as a function of applying PFG sequences and varying the external magnetic field strength result in distortion of $D_{app}(\Delta)$. Therefore, great care should be taken in interpreting the measured $D_{app}(\Delta)$ of ions/molecules in magnetically heterogeneous systems for the estimation of pore structures such as pore size and tortuosity.

The steady-state diffusion coefficients $D_{app,\infty}$ obtained at $\Delta = 90$ and 500 ms for APFG and MPFG, respectively, under an external magnetic field of 14.1 T (Figure 5) are in good

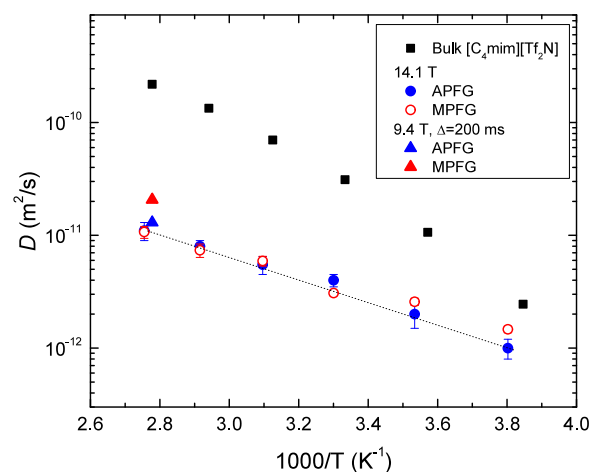


Figure 5. Diffusion coefficient of $[C_4mim]^+$ of bulk $[C_4mim][Tf_2N]$ (solid square) and the steady-state diffusion coefficient, $D_{app,\infty}$ of $[C_4mim]^+$ of $[C_4mim][Tf_2N]$, confined in ordered mesoporous carbon as a function of temperature measured by 1H PFG-NMR, using APFG and MPFG. Triangle symbols measured at $\Delta = 200$ ms and 9.4 T. The line is a guide for the eye.

agreement. Data obtained using APFG and MPFG at $\Delta = 200$ ms and 360 K under 9.4 T (on a 400 MHz NMR spectrometer with a maximum gradient strength of 50 G/cm) are also in good agreement. Because the MPFG needs longer Δ than APFG to overcome the internal gradient artifact (~ 6 times under 14.1 T), the $D_{app,\infty}$ from MPFG is a little larger than that from APFG. The $D_{app,\infty}$ represented in Figure 5 is to be considered the steady-state diffusion coefficient $D_{app,\infty}$ of the $[C_4mim]^+$ cation of $[C_4mim][Tf_2N]$ confined in OMC. These values are smaller by an order of magnitude than the bulk diffusion coefficient (D_0) in the high-temperature regime and moved toward convergence with D_0 as temperatures were decreased. This behavior is quite similar to the diffusion of the $[C_4mim]^+$ cation of $[C_4mim]-[Tf_2N]$ confined in mesoporous silica.^{9,15} Details of the dynamic and structural properties of $[C_4mim][Tf_2N]$ confined in mesoporous silica and carbon can be found in ref 15. Based on the Stokes–Einstein theory of diffusion, $D = k_B T / 6\pi\eta r_s$, where k_B is the Boltzmann constant, T is the absolute temperature, η is the viscosity, and r_s is the hydrodynamic radius of the diffusing molecules; D_∞ will be linearly proportional to T/η in the 3D bulk diffusion, as shown in Figure S6. This is consistent with values obtained from $[C_4mim]^+$ cations of $[C_4mim][Tf_2N]$ confined in mesoporous silica.⁹ The diffusion coefficients determined by MD simulations¹⁵ are slightly higher than those obtained by PFG-NMR in this work (Figure S7). The model construct for the MD simulations included ideal mesopores but not micropores, whose absence in the model yielded results that did not incorporate diffusion retardation that arises from the presence of micropores. However, the D_∞ determined by 1H PFG-NMR in this work is still in reasonable agreement with the calculated value obtained from the MD simulations.

Based on these observations, the diffusion behaviors observed by PFG-NMR showing $D_{app}(\Delta)$ (and consequent MSD) interrupted possibly due to the g_i artifact may need additional measurements using other PFG sequences or under different magnetic field strengths to confirm the presence of an effective g_i artifact on the measurement. Recently, it tended to increase by applying PFG-NMR to the electrolyte component diffusing within the carbon electrode.^{34,35} Then, as suggested here, the

internal gradient artifact on the PFG-NMR measurement for the fluid confined in the carbon pores should be confirmed before analyzing $D_{\text{app}}(\Delta)$, as Elwinger et al. tested the diffusion coefficient of the fluid confined in the pore glasses under two different magnetic fields of 7.05 and 11.75 T.³⁶

CONCLUSIONS

The apparent diffusion coefficients, $D_{\text{app}}(\Delta)$ of the cation $[\text{C}_4\text{mim}]^+$ of $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ confined in ordered mesoporous carbon, were measured with three different stimulated echo sequences, STE, APFG, and MPFG under two external magnetic fields of 9.4 and 14.1 T. It was observed in STE experiments that temperature-independent and magnetic field-dependent $D_{\text{app}}(\Delta)$ are due to the unsuppressed internal gradient artifact owing to the temperature-dependent paramagnetic susceptibility of OMC. The MPFG and APFG sequences run at 9.4 and 14.1 T yielded the same steady-state diffusion coefficients, while the early part of $D_{\text{app}}(\Delta)$ does not overlap. This suggests that PFG-NMR using symmetric (APFG) and asymmetric (MPFG) bipolar gradients can suppress the internal gradient artifacts at a long diffusion time. The MSD calculated from the $D_{\text{app}}(\Delta)$ varied as a function of the PFG sequences due to the differences in the effective internal gradient artifact. Therefore, caution should be exercised in interpreting the diffusion motion of ions/molecules in the pores from the MSD, especially species in pores with an inhomogeneous magnetic susceptibility from the pore surface to the pore volume. The $D_{\text{app}}(\Delta)$ showed that the diffusion of the $[\text{C}_4\text{mim}]^+$ cation confined in OMC is an order of magnitude smaller than that of bulk $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ but moved closer to the bulk diffusion coefficient as the temperature decreased. The MSD calculated from $D_{\text{app}}(\Delta)$ obtained by APFG and diffusion coefficients proportional to η/T suggest that the ionic liquid $[\text{C}_4\text{mim}][\text{Tf}_2\text{N}]$ diffuses like a 3D bulk liquid in the pores of OMC. The results demonstrated that it is useful to compare the $D_{\text{app}}(\Delta)$ obtained by using several PFG pulse sequences and/or under multiple external magnetic field strengths for determining accurate diffusion behavior of the ions and molecules interacting with magnetically inhomogeneous materials.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcc.5c08558>.

The Supporting Information is available free of charge on the ACS Publications Web site at DOI:

Magnetic susceptibility of mesoporous silica (KIT-6), apparent diffusion coefficients, and MSD of ILs confined in OMC and silica (PDF)

AUTHOR INFORMATION

Corresponding Author

Kee Sung Han – Physical and Computational Sciences Directorate, Pacific Northwest National Laboratory, Richland, Washington 99352, United States; Joint Center for Energy Storage Research (JCESR), Lemont, Illinois 60439, United States; orcid.org/0000-0002-3535-1818; Email: keesung.han@pnnl.gov

Author

Pasquale F. Pulvio – Chemical Sciences Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acs.jpcc.5c08558>

Notes

The authors declare no competing financial interest.

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